Dr. Abderraouf Maoudj

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Citations 1470+ h-index 13 i10-index 14

About

Dr. Abderraouf Maoudj is a senior robotics researcher specializing in next-generation robotic manipulation and learning systems. With more than a decade of combined academic and industrial R&D experience across Denmark, Algeria, and Saudi Arabia, he translates rigorous theory into field-tested robotic platforms in challenging real-world environments.

Currently a Post-Doctoral Research Fellow at the Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM, he conducts research on Next-Generation Robotic Manipulation and Learning Systems, where robots are not just our helpers but our skilled apprentices. His work focuses on developing robots that can act as adaptive learners and collaborative partners rather than simply replacing human labor. His research aims to:

- Acquire tool-usage skills directly by observing human actions.
- Learn new manipulation skills from minimal human/video demonstrations through vision–language–action (VLA) models.
- Adapt policies entirely offline using learned world models that simulate possible interactions. This approach leverages Recurrent State-Space Models (RSSM) to refine diffusion-based robotic skills through reinforcement learning, enabling safe and sample-efficient improvement while overcoming the generalization limits of imitation learning.
- Enable multi-cobot systems to coordinate and work as a team when tasks exceed individual capabilities.

Earlier roles include a three-year postdoctoral position at the University of Southern Denmark and seven years (Senior Researcher) at Algeria's Centre for the Development of Advanced Technologies (CDTA), where Dr. Abderraouf consistently translated academic breakthroughs into deployable, factory-floor solutions. At both SDU and CDTA, his research focused on enabling large teams of autonomous robots to coordinate and complement each other to perform complex tasks through localized decision-making. By leveraging distributed control and continuous learning, he has developed advanced methods to optimize multi-robot fleet coordination in warehouses and fulfillment centers, critical infrastructure that forms the backbone of modern e-commerce and logistics. His work has a significant impact on automated warehousing and smart manufacturing, enabling robust and adaptable systems for next-generation industrial automation.

Dr. Abderraouf has led and been involved in projects with industrial partners across Denmark, Algeria, and Saudi Arabia—including Siemens, Enable Robotics, Mobile Industrial Robots (MiR), and Sonacome, delivering AI-enabled solutions for manufacturing, logistics, and inspection.

Education

2013 – 2018	**Ph.D. Process Control & Robotics**, University of Sciences and Technology Houari Boumediene (USTHB). *Thesis:* Remote Control of a Heterogeneous Multi-Robot System: application to the RobuTER/ULM and AGV-M6 platforms.
2010 – 2012	**M.Sc. Electronics & Control**, Mohamed Seddik Ben Yahia University, Jijel. *Thesis:* Design and Realisation of a Mobile Robot.

2007 – 2010	**B.Sc. Electrical Engineering**, Mohamed Seddik Ben Yahia University, Jijel. *Capstone:* Development of a Card Reader.

Professional Experience

2023–Present	**Post-Doctoral Fellow**, Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM. Adaptive learning for next-generation robotic manipulation and learning systems, Collaborative robotics framework with closed-loop control, robust AI-based skill learning for collaborative operations in smart manufacturing, and AI-based methods for high-throughput coordination of autonomous mobile robot fleets moving materials in dense warehouses and fulfillment centers, critical infrastructures that form the backbone of modern e-commerce and logistics in KSA.
2021–2023	**Post-Doctoral Researcher**, SDU Biorobotics, Denmark. Focused on AI-based distributed multi-robot coordination for large-scale e-commerce logistics.
2014–2020	**Senior Researcher**, Robotics Division, CDTA, Algeria. Directed research and development of smart automation solutions tailored for industrial manufacturing, with emphasis on scalable integration of robotics and intelligent systems.

Selected R&D Projects

Ongoing Project	**Next-Generation Robotic Manipulation and Learning Systems for Saudi Industry 4.0** — Post-doctoral Researcher, IRC-IMR, KFUPM. This research aims to develop and demonstrate next-generation robotic manipulation systems that fundamentally transform how robots acquire and execute skills. My vision is to enable robots to understand human intent through natural communication, take plain instructions, learn tasks from minimal demonstrations, and coordinate multi-robot teams when tasks exceed individual capabilities. The work combines generative reinforcement learning, large vision–language–action models for grounding instructions, world models for safe offline skill refinement, and distributed intelligence for collaborative robot coordination.

2023 – 2025 **Integrating Collaborative Robot to Enhance Industrial tasks** — Post-doc, KFUPM.

Led Project Grant No. INMR2403, which redefines how collaborative robots learn, share, and execute skills in various dynamic production settings. The aim is to build a scalable collaborative robotics network that supports industrial environments with a foundation of shared and adaptable skills, in alignment with Saudi Arabia's Industrial Vision 2030. Contributions include:

- **Multi-cobot coordination and motion control:** Developed several AI-driven methods, including multi-agent deep reinforcement learning, imitation learning, and transformer-based inter-arm attention, for learning decentralized multi-arm motion planning. New videos of my work are available on our lab's channel: video
- **Swarm Robotics Coordination in Smart Manufacturing:** Addressed large-scale warehouse automation challenges in e-commerce, developing AI-based coordination mechanisms for fleets of mobile robots handling storage, transport, and shipment preparation. These algorithms improve efficiency in logistics operations and advance the foundations of smart manufacturing in line with Saudi Arabia's Industry 4.0 initiative.

2021 – 2023 **Swarm Robotics for Industry 4.0** — Postdoctoral Fellow.

The Fourth Industrial Revolution transforms manufacturing through robotics, IoT, AI, and cloud computing. This project studies decentralized coordination between mobile robotics in I4.0 manufacturing and warehouse scenarios to achieve higher levels of autonomy, scalability, flexibility, and robustness than current solutions offer. In this project, we developed AI-based algorithms for high-throughput coordination of autonomous mobile robot fleets moving materials in dense warehouses and fulfillment centers, critical infrastructures that form the backbone of modern e-commerce and logistics. Key outcomes:

- *Capacitated multi-agent control/scheduling framework*: Aiming to boost efficiency in logistics, warehouses, and transportation sectors, I designed and implemented a fully decentralized capacitated multi-agent control and scheduling framework specifically targeting logistics optimization. This system enables each autonomous guided vehicle (AGV) to reason locally while achieving fleet-level optimal solutions under capacity and compatibility constraints.
- *Swarm robots motion coordination in dynamic industrial environments and fulfillment centers*: Developed fast and robust decentralized planners coordinating up to thousands of robots in large dynamic warehouses. These research works improve order fulfillment efficiency, enhance operational efficiency, reduce costs, and strengthen global supply chain resilience.
- *Open-source contributions:* Released the `ANTS` planner and benchmark suite for research and teaching (GitHub, Benchmark dataset for lifelong MAPF, and video), now used in academia.

2014 – 2016 **Remote Control of Heterogeneous Robots** — Robotic Engineer|PhD student, CDTA.

As part of my doctoral research, this project involved:

- Developing a distributed multi-agent system for controlling industrial robot fleets.
- Implemented DRL-driven path optimization and fuzzy-logic controllers for mobile manipulators.
- Developing a fault-tolerant leader–follower control system for swarm mobile robots that maintains formation integrity under single-robot failures.

2016 – 2020 **Industry 4.0 Platform Development** — Senior Researcher, CDTA.

This project entailed designing and implementing advanced algorithms and control systems for Industry 4.0 platforms, seamlessly integrating robotic innovations with smart automation principles. Key responsibilities included:

- Integrated KUKA LBR iiwa with Java/ROS middleware for collaborative assembly.
- Implemented an RFID-based part-tracking system linked to Siemens PLCs, boosting inventory accuracy.
- Upgraded the facility's AGV fleet with a deep reinforcement learning navigation system to align with the control architecture of the I4.0 platform.

Teaching and Supervision

Teaching and mentorship have been integral to my academic journey. Over the years, I have conceived and delivered undergraduate and graduate courses, including *Introduction to Robotics and AI*, *Electronics for Robotics*, *PLC Automation*, *Autonomous Navigation*, and *Applied Robotics*—that blend simulator work and physical-robot labs. My dual experience in academia and industry enables me to lead research initiatives and ensure that students are well-prepared for the complexities of modern robotics applications, so that they gain problem-solving skills relevant to emerging industry needs.

Skills

Robotics: Python/C++, ROS 1/2, KUKA LBR iiwa, MiR, Enable Robotics MM, Meca500.

Control Electronics: Sensor conditioning; STM32/Arduino; Siemens TIA-Portal; real-time PLC control.

Software: Python, C/C++, Java, C#, MATLAB, R; Git, Docker, Linux.

Languages: Arabic (native), English (fluent), French (fluent).

Selected Publications

Journal Articles

1. A. Maoudj, A. Kouider, A. L. Christensen, "The capacitated multi-AGV scheduling problem with conflicting products: model and a decentralized multi-agent approach," *Robotics and Computer-Integrated Manufacturing*, 81:102514, 2023.
2. A. Maoudj, S. Mekid, M. Nouioua, "ML-BiRRT: Machine-learning-enhanced bi-directional RRT for multi-arm robot motion planning and coordination," *Robotics and Autonomous Systems*, under review, 2025.

3. A. Maoudj, A. L. Christensen, "Improved decentralized cooperative multi-agent path finding for robots with limited communication," *Swarm Intelligence*,18 (2), 167-185, 2024.
4. A. Maoudj, A. Hentout, "Optimal path planning based on a Q-learning algorithm for mobile robots," *Applied Soft Computing*, 97:106796, 2020.
5. U. Farhadi, H. Hess, A. Maoudj, A. L. Christensen, "Evolution of path costs for efficient decentralized multi-agent path finding," *Swarm and Evolutionary Computation*, 93:101833, 2025.
6. B. Mouad, A. Maoudj, A. L. Christensen, "HM-DRL: Enhancing multi-agent path finding with a heat-map-based heuristic for distributed deep reinforcement learning," *Applied Intelligence*, in press, 2025.
7. B. Mouad, A. L. Christensen, A. Maoudj, "MAPF-Relocate: Mitigating Congestion through Strategic Agent Relocation in Multi-Agent Pathfinding," *Autonomous Agents and Multi-Agent Systems*, 2025.
8. A. Kharidege, A. Maoudj, "Integration of a vision system with a robotic polishing system for machining a variety of workpieces," 2024.
9. A. Hentout, A. Maoudj, M. Aouache, "A review of fuzzy-logic approaches for collision-free path planning of manipulator robots," *Artificial Intelligence Review*, 56(4):3369–3444, 2023.
10. A. Hentout, A. Maoudj, A. Kouider, "Efficient fuzzy-logic control of indoor mobile robots evolving in static and dynamic workspaces," 2023.
11. A. Hentout, A. Maoudj, A. Kouider, M. Aouache, "Effective GA approach for municipal solid-waste collection: dynamic capacitated vehicle routing in smart cities," *International Journal of Artificial Intelligence and Soft Computing*, 7(4):329–352, 2022.
12. A. Hentout, M. Aouache, A. Maoudj, I. Akli, "Human–robot interaction in industrial collaborative robotics: a literature review of the decade 2008–2017," *Advanced Robotics*, 33(15–16):764–799, 2019.
13. A. Hentout, A. Maoudj, D. Guir *et al.*, "Collision-free path planning for indoor mobile robots based on rapidly-exploring random trees and PCHIP," *International Journal of Imaging and Robotics*, 19(3), 2019.
14. A. Hentout, A. Maoudj, M. Ouahal, A. Gasmi, "Fault-tolerant multi-agent scheme for leader/follower formation control of homogeneous mobile-robot teams," *International Journal of Systems, Control and Communications*, 10(2):95–125, 2019.
15. A. Hentout, A. Maoudj, D. Yahiaoui, M. Aouache, "RRT-A*-BT approach for optimal collision-free path planning for mobile robots," *Algerian Journal of Signals and Systems*, 4(2):39–50, 2019.
16. A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent scheduling and control system for robotic flexible assembly cells," *Journal of Intelligent Manufacturing*, 30:1629–1644, 2019.
17. A. Maoudj, A. Hentout, B. Bouzouia, R. Toumi, "On-line fault-tolerant fuzzy-based path planning and obstacle avoidance for manipulator robots," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 26(5):809–838, 2018.
18. A. Bakdi, A. Hentout, H. Boutami, A. Maoudj, O. Hachour, B. Bouzouia, "Optimal path planning and execution for mobile robots using a genetic algorithm and adaptive fuzzy-logic control," *Robotics and Autonomous Systems*, 89:95–109, 2017.
19. A. Hentout, A. Maoudj, N. Kaid-Youcef, D. Hebib, B. Bouzouia, "Distributed multi-agent bidding-based approach for collaborative mapping of unknown indoor environments," *Journal of Intelligent Systems*, 29(1):84–99, 2017.

Refereed Conference Papers

1. K. Amara, W. Dib, A. O. Djekoune, A. Maoudj, N. Zenati, "AquaRob: an unmanned surface drone for coastal surveillance and environmental monitoring," in *Proc. 4th Int. Conf. on Embedded & Distributed Systems (EDiS)*, IEEE, 2024, pp. 193–198.
2. A. Maoudj, S. N. Mekid, A. L. Christensen, "Lifelong MAPF for shipment preparation in stochastic warehouses," in *Proc. IEEE CASE*, 2024, pp. 1006–1011.
3. A. Maoudj, A. L. Christensen, "Decentralized multi-agent path finding in dynamic warehouse

environments," in *Proc. 21st ICAR*, IEEE, 2023.
4. A. Maoudj, A. L. Christensen, "Decentralized MAPF in warehouse environments for fleets of robots with limited communication," in *Proc. Int. Conf. on Swarm Intelligence*, Springer, 2022, pp. 104–116.
5. M. Gaham, M. Kadri, A. Maoudj *et al.*, "Multi-agent control of an Industry 4.0 cyber-physical assembly cell," in *Nat. Conf. on Applied Computing and Smart Technologies*, 2021.
6. A. Maoudj, A. L. Christensen, "Q-learning-based navigation for mobile robots in continuous and dynamic environments," in *Proc. IEEE CASE*, 2021, pp. 1338–1345.
7. A. Maoudj, A. Hentout, A. L. Christensen, A. Kouider, "A fast constructive path-planning algorithm for mobile-robot navigation," in *Proc. IEEE ETFA*, 2021, pp. 1–8.
8. A. Hentout, A. Mustapha, A. Maoudj, I. Akli, "Key challenges and open issues of industrial collaborative robotics," in *Workshop on Human-Robot Interaction: from Service to Industry (HRI-SI)*, 2018.
9. A. Hentout, A. Tiberkak, A. Maoudj *et al.*, "Virtual pheromone-based object-searching in RFID-enabled cyber-physical robotic systems," in *Proc. ICASS*, 2018, pp. 1–7.
10. A. Hentout, A. Beghni, A. B. Nourine, A. Maoudj, B. Bouzouia, "Mobile-robot RBPF-SLAM with LMS sensor in indoor environments," in *Proc. ICEE-B*, 2017, pp. 1–6.
11. A. Hentout, A. Maoudj, B. Bouzouia, "A survey of development frameworks for robotics," in *Proc. ICMIC*, 2016, pp. 67–72.
12. A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent scheduling in multi-robot cells using priority rules and a genetic algorithm," in *Proc. IEEE IECON*, 2016, pp. 692–697.
13. A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, "Task allocation and scheduling in cooperative heterogeneous robot teams: simulation results," in *Proc. IEEE INDIN*, 2015, pp. 179–184.

Book Chapters

1. A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, "Multi-agent architecture for telerobotics of heterogeneous robots via negotiated task allocation," in *1re Journée d'Étude en Automatique et ses Applications (1JEAA14)*, 2014, pp. 38–43.

References

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